A

AMPERES

FAILURE LAB

Every robot failure, caught, explained and fixed.

- Live: a Viam xArm6 that plays chess



The event arm and board, from the table camera

THE PROBLEM

Robots fail quietly

01 **Small failures, all day**

A missed grasp, a slip, a collision stop, a move logged that never happened.

02 **Evidence in four places**

Camera, gripper, arm controller and the operator's memory. Nobody lines them up.

03 **Found too late**

In the dataset, in the demo, or in front of the customer.

A REAL CASE FROM THIS ARM, 29 OCT

Arm	Motion complete	●
Gripper sensor	Holding a piece	●
Wrist camera	Pawn still on its square	●

Lab verdict: **PIECE_NOT_MOVED**

Each signal looks fine alone. Side by side they disagree: the move never happened.

Amperes Failure Lab, from failure to fix

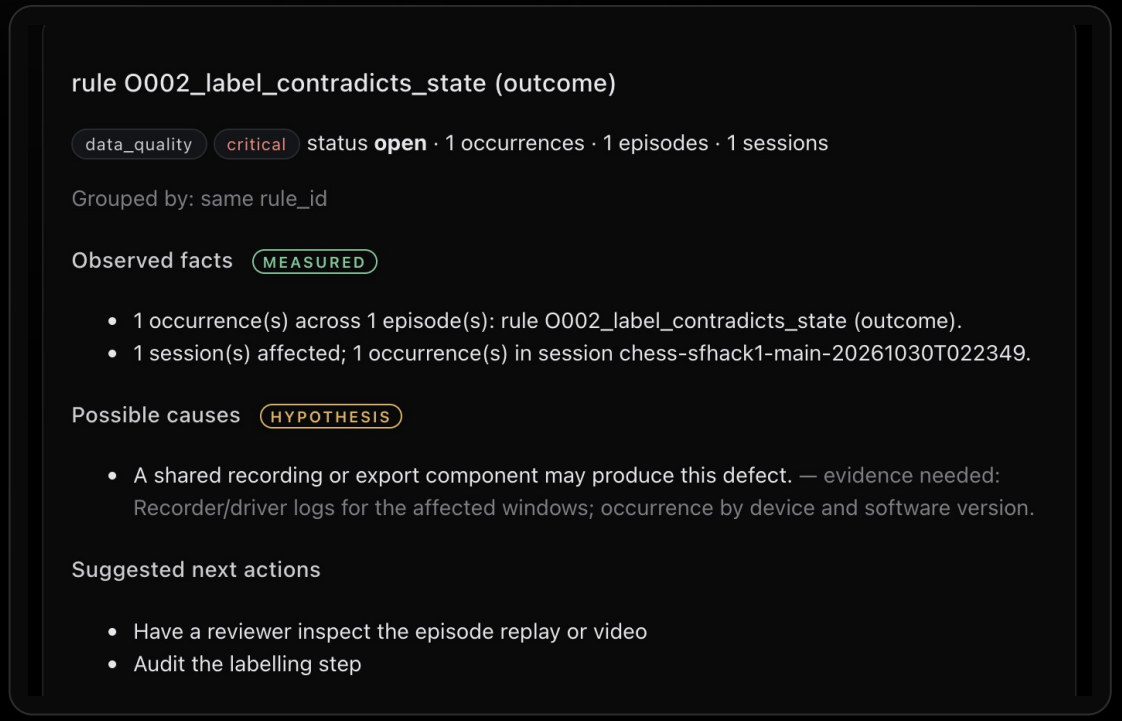
1. **Record**
   Images, depth, poses, gripper reading, outcome.
2. **Label**
   Missed grasp, slip, collision stop, refused motion.
3. **Triage**
   Recurring failures grouped, traced to a root cause.
4. **Verify**
   Replay and regression show the fix really helped.
5. **Export**
   Clean, labelled episodes as LeRobot datasets.

193 chess trials since 29 Oct

142 succeeded

38 failures labelled

13 invalid, kept out of training data



Incidents tab: a critical finding, measured facts kept apart from hypotheses

Local-first. The SQLite store stays on this machine; only opt-in Atlas sync and Claude piece recognition send data out.

Chess is an honest test for pick-and-place

32 pieces, 6 shapes

Small parts that look alike from straight above.

Millimetre placement

A few millimetres off and the gripper touches a neighbour.

A person at the board

Human hands work right next to the arm, every game.

Every action checkable

The rules of chess and two cameras say whether each move really happened.



The robot's own overhead view: every square located and labelled

If the lab can debug this in two days, it can debug your pick-and-place cell.

How the robot and the lab fit together

HARDWARE

UFactory xArm6
Six-axis arm

G2 gripper
Reports its finger position

RealSense, wrist
Colour + depth over the board

RealSense, tripod
Second view of the table

VIAM

Machine sfhack1
Arm, gripper and cameras picked by name

Motion service
Routed moves, not straight lines

Modelled obstacles
Table, walls and ceiling in the plan

DECISIONS

Stockfish
Picks every move

Claude vision
Names each piece

Depth heights
Check every name

Chess session
Verifies, retries, holds for hands

FAILURE LAB

Every move a trial
Images, depth, poses, gripper reading, outcome

Local store
SQLite under amperes_data/

Console
Label, triage, replay, export

Commands and telemetry flow through Viam; every result lands in the lab with its evidence.

THE PROOF, 29–30 OCT

What the lab caught on the real arm

CAMERA BIAS

~15 mm

before: grasping air beside pieces

4.5 mm

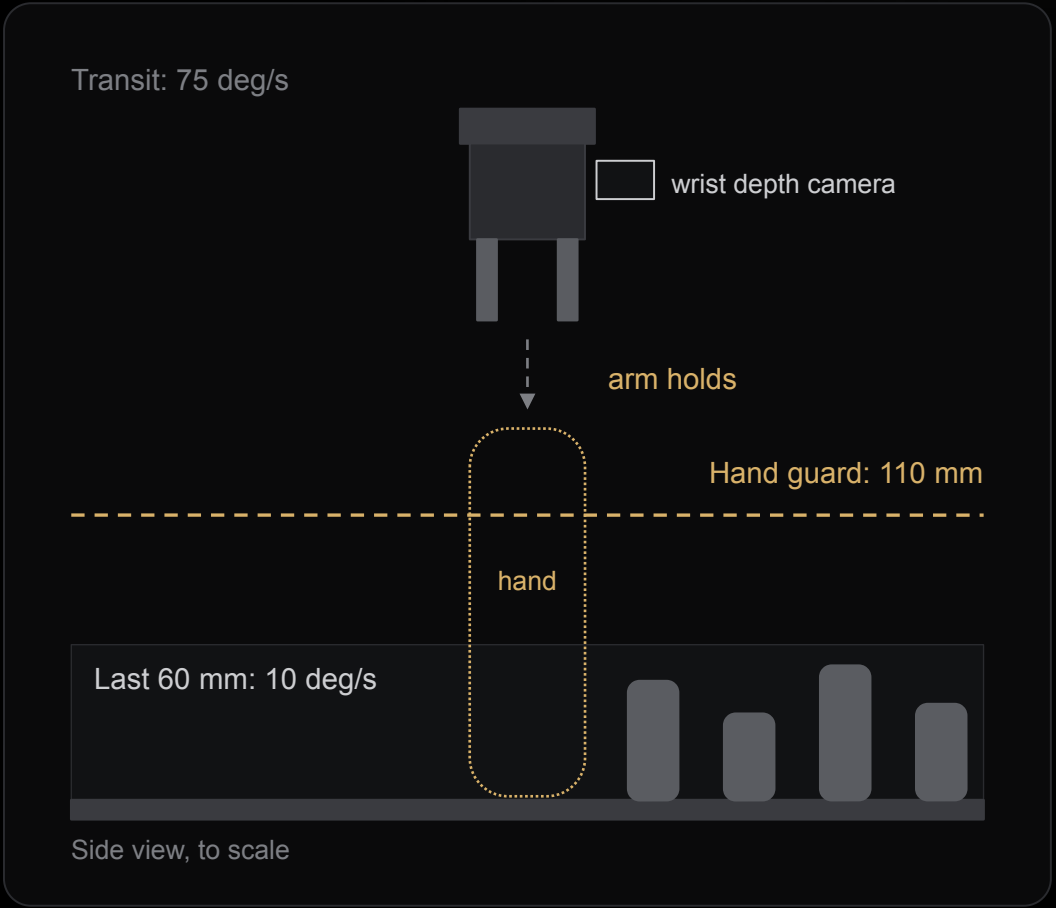
mean survey error after a fitted correction

14 / 14

pick-and-places in a row

FAILURE CAUGHT	FIX
Grasping air beside pieces	Camera bias measured on pieces the arm placed itself; fitted correction
Retried grips pressed pieces into the board	Place height follows the real grip; last 60 mm at 10 deg/s, transit 75 deg/s
Moves recorded that never happened	Empty-grip check, instant retries at small offsets, camera must confirm
Collision stops on long, fast swings	Big wrist turns split into two half-turns; motion routed in shorter moves
Board-reset planner looping	Loop guard: remembered states, forced follow-up moves, an operation limit
Typed move ignored while the camera watched	The watch now ends as soon as a move is entered
Pieces mislabelled (a rook read as a pawn)	Claude vision, height check, confidence gate, corrections that follow the piece

Safe around people and pieces



Hand guard

The wrist depth camera looks under the gripper before each descent. Anything over 110 mm up: hold.

Safety envelope

Every target checked against clearance margins; no orientation change above 100 degrees in one move.

Gentle near the pieces

Transit at 75 deg/s; the last 60 mm above the board at 10 deg/s.

Telemetry watchdog

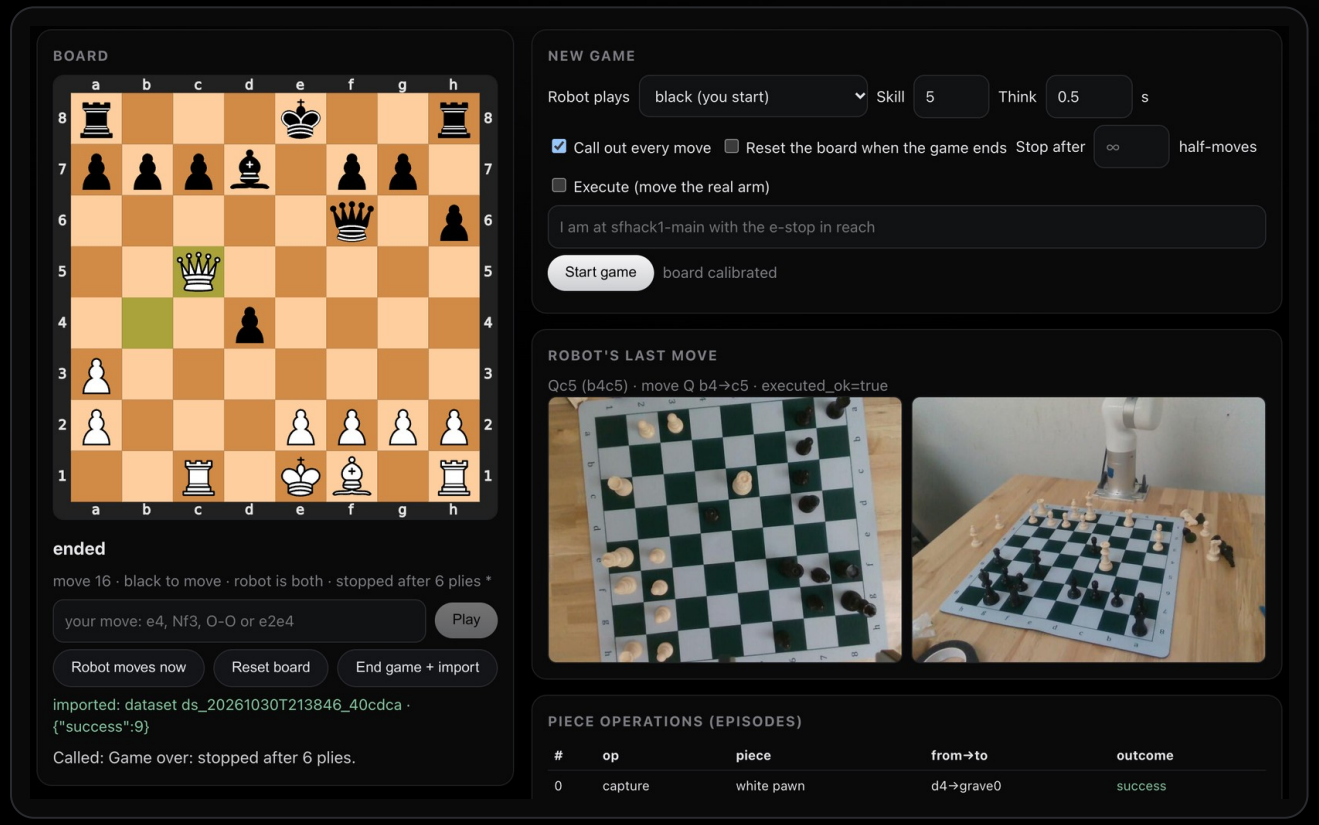
If arm telemetry stops during motion, the arm stops.

Operator in the loop

Nothing moves until the operator types the acknowledgement, e-stop in reach. STOP works mid-move.

LIVE DEMO

Robot vs robot, every move called out loud



CALLED OUT AFTER EVERY MOVE

“White knight to F3, check.”

6 / 6 moves confirmed by the wrist camera and called out loud in the 2:33 PM rehearsal

Only real moves count

Gripper and camera must agree; retries at once.

Every move is a trial

Lands in the Failure Lab when the game ends.

Or play it yourself

Move a piece and the wrist camera sees it.

REUSE

Built to reuse on your robot

132 tests passing

Any Viam robot

Arm, gripper, motion and cameras are chosen by name, so any Viam arm, gripper and camera works.

```
AMPERES_ARM  AMPERES_GRIPPER
AMPERES_MOTION  AMPERES_CAMERAS
```

Any recorded session

The lab ingests recorded Viam sessions, JSONL and CSV packages, and LeRobot datasets.

```
amperes/ingest
jsonl  csv_package  lerobot_v3
```

Clean datasets out

Labelled episodes export as LeRobot datasets, depth included. Failed attempts stay in, labelled.

```
amperes/export_lerobot.py
```

Fleet memory on MongoDB Atlas

Opt-in. Atlas Vector Search finds the same failure across sessions: a missed grasp from 30 Oct matched 6 others with identical recorded evidence, one from the evening before. Signatures are deterministic, built from that evidence.

```
python -m amperes.cli atlas sync <dataset_id>
python -m amperes.cli atlas similar <episode_id> -k 6
```

HONEST LIMITS

- Cannot stand a fallen piece back up.
- Piece names from above sometimes need an operator fix.
- Speed is capped by the arm's own collision detection.

AMPERES

FAILURE LAB

The lab is the product. The chess robot is the proof.

CODE

github.com/vnmoorthy/amperes-haunted-chess

LANDING PAGE

vnmoorthy.github.io/amperes-haunted-chess

- Live now: the arm playing both sides